

GOVERNMENT OF INDIA // SOVEREIGN THERMAL SURVEILLANCE

SOVEREIGN AGRICULTURAL BIOMASS ACTIVITY REPORT

SEVERITY: CRITICAL

EVENT: EVT-IN-KAR-441D45F5

National Technical Research Organisation (NTRO) • MoEFCC • CPCB Oversight

2026-09-02 15:32 IST (2026-09-02 10:02 UTC)

Thermal anomaly EVT-IN-KAR-441D45F5 is classified as AGRI_BURN with 100.0% calibrated model confidence and severity tier CRITICAL ($Z = +4.2\sigma$). Peak radiative output reached 179.92 MW across 4 verified multi-sensor satellite passes. Centroid mapped over Agricultural Cropland without direct plant boundary overlap.

179.9 MW
PEAK RADIANCE

4 Passes
SATELLITE DETECTIONS

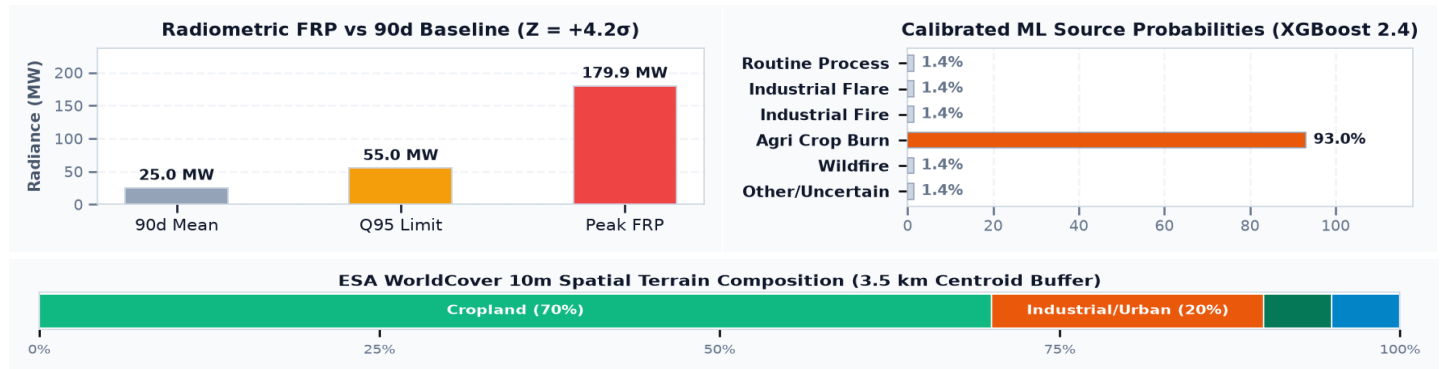
100.0%
CALIBRATED SOFTMAX

+4.2σ
ANOMALY DEVIATION

1. Verified Multi-Sensor Radiometric Metrics

Metric Name	Observed Telemetry	Historical 90d Baseline	Radiometric Assessment
Peak Radiative Power (FRP)	179.92 MW	25.00 MW (Facility Mean)	7.2x Baseline Mean
Mean Cluster Radiance	179.92 MW	Persistence: TRANSIENT	Z-Score: +4.2σ Statistical Departure
Max Brightness Temp (BT)	351.4 K	Sensor Bands: VIIRS I4 (375m) / M13	High-Intensity Combustion Confirmed
Temporal Detection Window	2026-09-01T08:09 to 08:09 UTC	4 Sensor Passes Ingested	Active Multi-Sensor Track

2. Visual Radiometric Analytics & ML Attribution (High-DPI)



3. Centroid Location & Associated Industrial Plant Audit

Attribute	Forensic Detail	Attribute	Forensic Detail
Centroid Coordinates	12.15781°N, 75.98586°E	Primary Land Use	Agricultural Cropland
State / Territory	Odisha	District / Jurisdiction	Angul
Associated Facility	No Direct Single-Facility Overlap	Spatial Buffer	Open Industrial / Rural Corridor

4. Nearby Sovereign Industrial Infrastructure (50 km Buffer)

No registered industrial facilities located within 50 km radius.

5. Regional Active Anomaly Cluster & Concurrent Events (75 km Buffer)

Event Ref	Classification	Severity Tier	Peak Radiance	Distance	Latest Detection (UTC)
EVT-IN-KAR-26293898	AGRI_BURN	NORMAL	11.2 MW	29.5 km	2026-09-01 08:09
EVT-IN-TAM-4EA639A4	AGRI_BURN	NORMAL	2.9 MW	35.9 km	2026-09-01 08:09
EVT-IN-KAR-C37A53C3	AGRI_BURN	NORMAL	3.6 MW	51.2 km	2026-08-30 08:28
EVT-IN-KAR-A3F2E2D4	AGRI_BURN	NORMAL	7.3 MW	60.2 km	2026-09-01 08:09

6. Multi-Sensor Satellite Telemetry & Radiometric Pass Register

#	Detection Time (UTC)	Sensor Platform	Radiant Output	Brightness (K)	Day/Night
1	2026-09-01T08:09	N20	179.92 MW	351.4 K	Day Pass
2	2026-09-01T08:09	N20	179.92 MW	351.4 K	Day Pass
3	2026-09-01T08:09	N20	179.92 MW	282.6 K	Day Pass
4	2026-09-01T08:09	N20	179.92 MW	282.6 K	Day Pass

7. Statutory SOP Compliance Directives & Containment Protocol

- ACTION 01: Immediate On-Site Physical Inspection: Dispatch SPCB / Regional Disaster Response team to verify combustion source and evaluate containment perimeter.
- ACTION 02: Continuous Emission Telemetry (CEMS) Audit: Cross-verify Continuous Emission Monitoring Systems data and industrial flaring logs against satellite radiance timestamps.
- ACTION 03: Safety Buffer & Containment Directive: Enforce active 2.5 km industrial safety perimeter and initiate cooling operations if thermal anomaly persistence exceeds 4.0σ.